WALMART

Sales Analysis

Retail Data Cleaning, SQL Analysis & Power BI Dashboard

PROJECT PURPOSE

Transform a retail sales dataset into a cleaned, queryable dataset and an interactive Power BI dashboard that helps explore sales performance, profitability, product performance, and regional differences.

Prepared as a data analytics portfolio project

Tools: Excel • MySQL • Power BI

1. Project Overview

This project demonstrates an end-to-end analytics workflow: inspect and clean a retail dataset in Excel, validate and query the prepared data in MySQL, and present key metrics in an interactive Power BI report.

Item	Description
Project name	Walmart Sales Analysis
Domain	Retail / sales analytics
Tools	Excel for data inspection and preparation; MySQL for validation and SQL analysis; Power BI Desktop for dashboarding
Primary outputs	Excel-prepared dataset, cleaned CSV, SQL analysis queries, Power BI report (.pbix), this documentation

2. Objectives

- Prepare a consistent dataset for analysis by checking field formats, missing values, and date parsing.
- Use SQL aggregation and ranking queries to explore sales, profit, quantity, products, categories, and locations.
- Build a Power BI dashboard with headline KPIs and four focused visuals.
- Communicate findings clearly while avoiding unsupported conclusions that are not backed by calculated results.

3. Dataset Description

The project uses a retail sales dataset downloaded from Kaggle. Excel was used to inspect and prepare the source data, including splitting the original Geography information into separate country, city, and state fields. The prepared data was then loaded into MySQL for validation and analysis and exported as a UTF-8 CSV for Power BI. The project is named Walmart Sales Analysis; describe the dataset as a retail sales dataset unless its Walmart provenance has been independently verified.

Column	Meaning / intended use
state	State/region used for geographic comparisons
ship_date	Shipping date; used for delivery/shipping-related time checks if needed
sales	Sales amount; aggregated for revenue/sales performance
quantity	Quantity sold; aggregated to measure volume
profit	Profit amount; used to assess profitability and losses
product_name	Product description used for product ranking
order_id	Order identifier; distinct count used as an order KPI
order_date	Date the order was placed; used for monthly trend and date slicer
email_id	Email/customer-related identifier present in source data; handle carefully and do not expose in public dashboard
country	Country field
city	City used for local/regional comparisons
category	Product category used for category analysis and filtering

Data Import and Row Count Validation:

The raw Excel file and cleaned CSV file may contain different numbers of rows due to data-cleaning operations or issues during the import process. Therefore, the row counts were identified as an important validation point before interpreting the analysis results. The cleaning steps and MySQL import process should be verified to determine whether any records were removed, skipped, or rejected unintentionally. The total number of rows in MySQL should be compared with the intended source file to ensure data consistency and completeness. Any import warnings or rejected records should also be investigated to identify potential data loss and improve the reliability of the analysis.

4. Data Preparation and Cleaning

Excel was used for initial inspection and preparation, while MySQL was used for date parsing, data-quality checks, aggregations, and business analysis. The documented preparation workflow included these checks and transformations:

- Reviewed the dataset columns and standardized the table structure for analysis.
- Split the original Geography information into separate country, city, and state fields, then removed the combined Geography field.
- Checked for null or blank values across the analysis columns; the checks reported no missing/blank values in the working dataset at that stage.
- Reviewed numeric ranges for sales, quantity, and profit. The working data showed sales from 0.99 to 13,999.96, quantity from 1 to 14, and profit from -3,399.98 to 6,719.98.
- Reviewed apparent repeated combinations of order_id, product_name, and order_date. Rows with differing numeric values were not automatically deleted because they may represent distinct line items.
- Converted date strings from DD-MM-YYYY into true date values for analysis/export. The source date format used hyphens, so the matching MySQL format string was %d-%m-%Y.
- Exported the cleaned query result to CSV using UTF-8 encoding for import into Power BI.

5. Excel Workflow

Excel was used as the first preparation stage before SQL analysis. The workflow focused on making the columns consistent and preparing a clean, structured file for database import.

5.1 Inspect and clean the source data

- Open the raw CSV/XLSX and review the headers, row count, and sample records.
- Check for blank cells and inconsistent data formats in the fields needed for analysis.
- Split the Geography field into country, city, and state where appropriate, verify the split results, then remove the original combined field once the separate columns are correct.
- Confirm that sales and profit are numeric and quantity is a whole-number field.
- Review order_date and ship_date formats. The working source used DD-MM-YYYY strings; avoid converting ambiguous dates without checking the intended day/month order.
- Export the prepared file as CSV UTF-8 so text is preserved when importing into MySQL and Power BI.

5.2 Excel quality checks

- Compare the row count before and after cleaning to make sure rows were not accidentally dropped.
- Use filters or conditional formatting to locate blanks and inconsistent values.
- Check numeric columns for unexpected text values, negative values, or unusually high/low values; investigate rather than automatically deleting valid business cases.
- Keep an untouched copy of the raw file and save the cleaned file separately.

6. Business Questions and SQL Analysis

The SQL analysis is designed to answer practical questions about overall performance, categories, products, geography, time trends, and profitability.

Question 1: What is the overall business performance?

total_records	total_sales	total_profit	total_quantity_sold	average_sales_per_record
3099	713471.45	106021.47	11843	230.23

Question 2: Which product categories generate the most sales and profit?

category	total_sales	total_profit	total_quantity_sold
Chairs	101781.36	4027.62	734
Phones	98588.49	9078.69	1054
Tables	84754.6	1482.54	481
Storage	66569.23	8657.98	986
Accessories	61114.1	16484.62	1032
Binders	54456.74	15633.66	1758
Copiers	49749.23	19327.25	88
Machines	42444.14	-618.95	147
Bookcases	36004.18	-1646.5	306
Appliances	30236.32	8261.29	492
Furnishings	25910.22	6724.72	1111
Paper	25235.58	11469.65	1561
Supplies	17905.32	564.54	220
Art	9212.06	2374.15	926
Labels	5078.72	2303.07	480
Envelopes	3520.36	1627.78	214
Fasteners	910.8	269.36	253

Question 3: What are the top 10 products by sales?

product_name	total_sales	total_profit	total_quantity_sold
Canon imageCLASS 2200 Advanced Copier	13999.96	6719.98	4
High Speed Automatic Electric Letter Opener	13100.24	524.01	8
Global Troy Executive Leather Low-Back Tilter	10019.61	626.24	25
Fellowes PB500 Electric Punch Plastic Comb Bind...	8134.33	3050.38	8
GuestStacker Chair with Chrome Finish Legs	8030.01	802.99	27
Okidata MB760 Printer	7834.4	881.37	7
Bretford Rectangular Conference Table Tops	7710.66	180.53	26
LogitechÂ P710e Mobile Speakerphone	7467.21	1418.78	29
Canon PC1060 Personal Laser Copier	6719.9	2267.97	12
Hewlett Packard LaserJet 3310 Copier	6239.89	2183.96	13

Question 4: Which States Generate the Highest Sales and Profit?

state	total_sales	total_profit	total_quantity_sold
California	450567.63	74669.41	7426
Washington	136590.2	32976.67	1809
Arizona	34284.92	-3439.27	838
Colorado	31285.25	-6489.68	679
Oregon	17282.09	-1206.79	475
Nevada	16642.9	3291.76	163
Utah	10575.04	2354.87	187
Montana	5583.25	1831.11	54
New Mexico	4783.54	1157.13	151
Idaho	4273.49	776.06	57
Wyoming	1603.14	100.2	4

Question 5: Which Cities Generate the Highest Sales?

city	state	total_sales	total_profit
Los Angeles	California	173168.85	29807.11
Seattle	Washington	117772.59	28869.03
San Francisco	California	110917.03	17176.67
San Diego	California	47115.08	6309.35
Denver	Colorado	11690.08	-244.91
Phoenix	Arizona	10815.15	-2796.49
North Las Vegas	Nevada	9801	2247.68
Anaheim	California	7955.9	1220.04
Fresno	California	7861.03	938.98
Sacramento	California	7299.48	2065.71

Question 6: What Is the Monthly Sales Trend?

sales_year	sales_month	total_sales	total_profit
2011	1	2648.05	20.97
2011	2	1281.5	201.63
2011	3	11008.91	1534.24
2011	4	9259.63	1599.96
2011	5	6570.44	873.57
2011	6	9629.41	1771.84
2011	7	21605.75	1875.63
2011	8	12967.96	2555.93
2011	9	14669.69	2972.47
2011	10	8719.34	1387.87

Question 7. Which Products Generate the Highest Profit?

product_name	total_sales	total_profit	total_quantity_sold
Canon imageCLASS 2200 Advanced Copier	13999.96	6719.98	4
Fellowes PB500 Electric Punch Plastic Comb Bind...	8134.33	3050.38	8
Canon PC1060 Personal Laser Copier	6719.9	2267.97	12
Hewlett Packard LaserJet 3310 Copier	6239.89	2183.96	13
LogitechÂ Z-906 Speaker sys - home theater - 5...	4289.87	1715.95	13
Plantronics Savi W720 Multi-Device Wireless He...	3797.55	1670.92	9
Ibico EPK-21 Electric Binding System	4535.98	1644.29	3
Canon PC940 Copier	3149.93	1480.47	7
LogitechÂ P710e Mobile Speakerphone	7467.21	1418.78	29
Canon Image Class D660 Copier	2999.95	1379.98	5

Question 8. Which Products Generate the Biggest Losses?

product_name	total_sales	total_profit
Lexmark MX611dhe Monochrome Laser Printer	2549.99	-3399.98
Zebra GK420t Direct Thermal/Thermal Transfer ...	703.71	-938.28
O'Sullivan 4-Shelf Bookcase in Odessa Pine	1808.65	-802.11
Hon 2090 Â"Pillow SoftÂ" Series Mid Back Swivel...	3371.76	-547.9
Atlantic Metals Mobile 4-Shelf Bookcases, Custo...	2261.9	-491.71
Bestar Classic Bookcase	1179.88	-439.94
Chromcraft 48" x 96"" Racetrack Double Pedest...	1314.62	-436.07
Bretford Â"Just In TimeÂ" Height-Adjustable Mu...	5008.8	-425.75
GBC DocuBind 300 Electric Binding Machine	1788.33	-399.74
Panasonic KX MC6040 Color Laser Multifunction ...	269.97	-386.96

Question 9. Which Categories Have the Highest Profit Margins?

category	total_sales	total_profit	profit_margin_percentage
Envelopes	3520.36	1627.78	46.24
Paper	25235.58	11469.65	45.45
Labels	5078.72	2303.07	45.35
Copiers	49749.23	19327.25	38.85
Fasteners	910.8	269.36	29.57
Binders	54456.74	15633.66	28.71
Appliances	30236.32	8261.29	27.32
Accessories	61114.1	16484.62	26.97
Furnishings	25910.22	6724.72	25.95
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Phones	98588.49	9078.69	9.21
Chairs	101781.36	4027.62	3.96
Supplies	17905.32	564.54	3.15
Tables	84754.6	1482.54	1.75
Machines	42444.14	-618.95	-1.46
Bookcases	36004.18	-1646.5	-4.57

Question 10. What Is the Overall Business Performance?

unique_orders	unique_products	unique_cities	unique_states	overall_profit_margin_percentage
1578	1448	168	1	14.86

7. Power BI Dashboard



8. Summary

Summary and Business Recommendations

This project demonstrates a retail sales analysis workflow using Excel, MySQL, and Power BI. The analysis evaluates sales performance, profitability, product and category performance, geographic sales distribution, and monthly trends. The Power BI dashboard presents key performance indicators and visual insights to support data-driven decision-making.

Based on the analysis, the following business recommendations can help improve sales performance and profitability:

1. **Improve Product Performance:** Identify top-selling products and maintain sufficient inventory to meet customer demand. Review low-performing products to determine whether pricing, promotion, or product assortment changes are needed.
2. **Optimize Profitability:** Evaluate products and categories with low or negative profits. Review pricing strategies, discounts, and operating costs to identify opportunities to improve profit margins.
3. **Focus on High-Performing Locations:** Identify states and cities generating strong sales and investigate opportunities to expand successful strategies to lower-performing locations.
4. **Plan for Seasonal Demand:** Use monthly sales trends to anticipate changes in demand, improve inventory planning, and schedule promotions during periods when sales are typically lower.
5. **Support Data-Driven Decisions:** Use the Power BI dashboard to monitor sales, profit, order volume, and quantity regularly, helping decision-makers identify performance changes and respond promptly.

Overall, this project demonstrates practical skills in data preparation, SQL analysis, business intelligence, and dashboard development. The findings can help businesses identify sales opportunities, monitor profitability, and make more informed operational decisions. Further validation of the dataset's row counts and source information is recommended to ensure the reliability of the results.